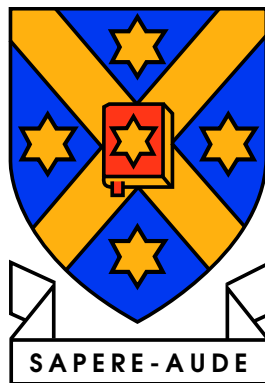


Thesis title here

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Abstract

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Acknowledgements

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Introduction

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Chapter 2

Background

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References

Rountree, N. (1998). *How to do Anything*. Dunedin, New Zealand: Non-existent Publishers.

Appendix A

Derivations

A.1 Magnetohydrodynamics (MHD)

Ideal MHD Equations

The ideal MHD equations encode the conservation laws for a conducting, flowing fluid. The continuity equation represents mass conservation. It is derived by arguing that the mass in a volume of fluid is equivalent to the mass flux in and out of the same volume.

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0. \quad (\text{A.1})$$

Similarly, the MHD equation for momentum conservation is obtained by equating the momentum change within a volume with processes that transport momentum in and out of the volume. The momentum due to viscous stress is negligible.

$$\rho \left(\frac{\partial}{\partial t} + \mathbf{u} \cdot \nabla \right) \mathbf{u} = -\nabla p + \frac{(\nabla \times \mathbf{B}) \times \mathbf{B}}{4\pi} = -\nabla \left[\left(p + \frac{B^2}{8\pi} \right) \mathbf{I} \right] + \frac{\mathbf{B} \cdot \nabla \mathbf{B}}{4\pi}. \quad (\text{A.2})$$

The momentum conservation equation contains the MHD body force, which is obtained by combining Maxwell's equations of electromagnetism and the Lorentz force, limited to the non-relativistic case.

$$\mathbf{F} = \frac{(\nabla \times \mathbf{B}) \times \mathbf{B}}{4\pi}.$$

By substituting the Ampere-Maxwell law and Ohm's law into Faraday's law and assuming no magnetic diffusivity, we reach the ideal induction equation. This MHD equation implies that the fluid elements will remain on the same magnetic field line. It also represents the conservation of flux since the field lines are 'frozen into the flow'.

Thus, changing the geometric properties of a fluid element can amplify magnetic fields in the fluid

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{u} \times \mathbf{B}) \quad (\text{A.3})$$

The adiabatic entropy equation for MHD is derived by applying the energy conservation law. The total energy inside a fluid element is composed of kinetic, internal and magnetic energies. The change in the total energy in a volume is equal to the energy transport processes and we have ignored negligible processes such as heat flux.

$$\left(\frac{\partial}{\partial t} + \mathbf{u} \cdot \nabla \right) s = 0 \quad (\text{A.4})$$

The variable s represents entropy and is defined as $s \equiv p/\rho^\gamma$.

A.2 MHD Waves

We can find solutions to the ideal MHD equations in the form of a background field and a perturbation.

$$\rho = \rho_0 + \delta\rho, \quad p = p_0 + \delta p, \quad \mathbf{B} = B_0 \hat{\mathbf{z}} + \delta \mathbf{B}, \quad \mathbf{u} = \mathbf{U} + \delta \mathbf{u}$$

The background velocity field is aligned in the direction of the magnetic field and varies along the perpendicular direction, $\mathbf{U} = U(x)\hat{\mathbf{z}}$, introducing a shear in the background velocity field. The perturbation in the velocity field is $\delta \mathbf{u} = \partial \boldsymbol{\xi} / \partial t$, where we have introduced the fluid displacement field $\boldsymbol{\xi}$.

By assuming the perturbations are infinitesimal we can consequently linearise the MHD equations.

Expanding the continuity equation, then substituting the solution for the density field, the following equations are obtained.

$$\begin{aligned} \frac{\partial \rho}{\partial t} + \delta \mathbf{u} \cdot \nabla \rho &= -\rho \nabla \cdot \delta \mathbf{u} \\ \frac{\partial}{\partial t}(\rho_0 + \delta \rho) + \frac{\partial \boldsymbol{\xi}}{\partial t} \cdot \nabla(\rho_0 + \delta \rho) &= -(\rho_0 + \delta \rho) \nabla \cdot \frac{\partial \boldsymbol{\xi}}{\partial t} \end{aligned}$$

We subsequently linearise the equation by ignoring the second order or higher perturbation terms and simplify. Note that the nonlinear fluid advection term $\delta \mathbf{u} \cdot \nabla$ is completely removed from the equation since it always produces a second order term.

$$\frac{\partial \delta \rho}{\partial t} = -\rho_0 \nabla \cdot \frac{\partial \boldsymbol{\xi}}{\partial t}$$

$$\frac{\delta\rho}{\rho_0} = -\nabla \cdot \boldsymbol{\xi} \quad (\text{A.5})$$

The physical interpretation of the equation above is that the contraction (negative divergence) of a fluid element increases its density.

Ignoring the nonlinear fluid advection term, we can evaluate the energy equation using the quotient rule.

$$\begin{aligned} \frac{\partial}{\partial t} \frac{p}{\rho^\gamma} &= 0 \\ \frac{\frac{\partial p}{\partial t} \rho^\gamma - \frac{p}{\rho} \gamma \rho^{\gamma-1}}{\rho^{2\gamma}} &= 0 \end{aligned}$$

Substituting the density and pressure fields and linearising, we obtain

$$\begin{aligned} \frac{\partial(p_0 + \delta p)}{\partial t} &= \frac{(p_0 + \delta p)}{(\rho_0 + \delta \rho)} \gamma \frac{\partial(\rho_0 + \delta \rho)}{\partial t} \\ \frac{\partial}{\partial t} \frac{\delta p}{p_0} &= \gamma \frac{\partial}{\partial t} \frac{\delta \rho}{\rho_0} \\ \frac{\delta p}{p_0} &= -\gamma \nabla \cdot \boldsymbol{\xi} \end{aligned} \quad (\text{A.6})$$

where we have used the result from the linearised continuity equation (A.5) to simplify the equation.

Substituting the perturbed solutions and removing the advection term in the induction equation leads to the following.

$$\begin{aligned} \left(\frac{\partial}{\partial t} + \delta \mathbf{u} \cdot \nabla \right) \mathbf{B} &= \mathbf{B} \cdot \nabla \delta \mathbf{u} - \mathbf{B} \nabla \cdot \delta \mathbf{u} \\ \frac{\partial}{\partial t} (B_0 \hat{\mathbf{z}} + \delta \mathbf{B}) &= (B_0 \hat{\mathbf{z}} + \delta \mathbf{B}) \cdot \nabla \frac{\partial \boldsymbol{\xi}}{\partial t} - (B_0 \hat{\mathbf{z}} + \delta \mathbf{B}) \nabla \cdot \frac{\partial \boldsymbol{\xi}}{\partial t} \end{aligned}$$

After linearization, we can split the spatial derivative ∇ and fields into a parallel projection, $\parallel$, in the direction of the background magnetic field $\hat{\mathbf{z}}$ and a perpendicular projection, $\perp$, in the x-y plane. Since the magnetic field lines are carried by the fluid flow, we note that the fluid displacement parallel to the magnetic field does not produce perturbations.

$$\begin{aligned} \frac{\partial \delta \mathbf{B}}{\partial t} &= B_0 \nabla_{\parallel} \frac{\partial \boldsymbol{\xi}}{\partial t} - \hat{\mathbf{z}} B_0 \nabla \cdot \frac{\partial \boldsymbol{\xi}}{\partial t} \\ \frac{\delta \mathbf{B}}{B_0} &= \nabla_{\parallel} \boldsymbol{\xi} - \hat{\mathbf{z}} \nabla \cdot \boldsymbol{\xi} \\ \frac{\delta \mathbf{B}_{\perp}}{B_0} + \frac{\delta B_{\parallel}}{B_0} &= \nabla_{\parallel} \boldsymbol{\xi}_{\perp} - \hat{\mathbf{z}} \nabla_{\perp} \cdot \boldsymbol{\xi}_{\perp} \end{aligned}$$

When we separate the magnetic field into its magnitude B and direction $\mathbf{b} = \mathbf{B}/B$, we see that for the perturbation $\delta\mathbf{B}$, $\delta\mathbf{b}$ must be perpendicular to $\hat{\mathbf{z}}$ in order for a perpendicular perturbation to exist.

$$\frac{\delta\mathbf{B}}{B_0} = \frac{\delta(B\mathbf{b})}{B_0} = \delta\mathbf{b} + \hat{\mathbf{z}}\frac{\delta B}{B_0}.$$

Hence, the perpendicular and parallel perturbations of the magnetic represent perturbations in the direction and strength of the field and can be expressed in terms of the perpendicular displacement field.

$$\begin{aligned}\frac{\delta\mathbf{B}_\perp}{B_0} &= \delta\mathbf{b}, & \frac{\delta B_\parallel}{B_0} &= \frac{\delta B}{B_0}.\end{aligned}$$

$$\frac{\delta\mathbf{B}_\perp}{B_0} = \nabla_\parallel \boldsymbol{\xi}_\perp, \quad \frac{\delta B_\parallel}{B_0} = -\nabla_\perp \cdot \boldsymbol{\xi}_\perp \quad (\text{A.7})$$

Finally, we remove the nonlinear term in the momentum equation to obtain

$$\rho \frac{\partial}{\partial t} \delta\mathbf{u} = -\nabla p - \nabla \frac{B^2}{8\pi} + \frac{\mathbf{B} \cdot \nabla \mathbf{B}}{4\pi}.$$

After linearisation, the left side of the equation becomes

$$(\rho_0 + \delta\rho) \frac{\partial}{\partial t} \left(\frac{\partial \boldsymbol{\xi}}{\partial t} \right) = \rho_0 \frac{\partial^2 \boldsymbol{\xi}}{\partial t^2}. \quad (\text{A.8})$$

The term ∇p resolves into the form below after the substitution of the linearised energy equation (A.6).

$$\begin{aligned}-\nabla(p_0 + \delta p) &= -\nabla \delta p \\ &= \gamma p_0 \nabla(\nabla \cdot \boldsymbol{\xi})\end{aligned} \quad (\text{A.9})$$

Linearising the last two terms in the momentum equation we get

$$\begin{aligned}-\frac{1}{8\pi} \nabla (B_0 \hat{\mathbf{z}} + \delta\mathbf{B})^2 &= -\frac{1}{8\pi} \nabla (2B_0 \hat{\mathbf{z}} \cdot \delta\mathbf{B}) \\ &= -\frac{B_0^2}{4\pi} \nabla \frac{\delta B}{B_0}\end{aligned}$$

and

$$\begin{aligned}\frac{1}{4\pi} (B_0 \hat{\mathbf{z}} + \delta\mathbf{B}) \cdot \nabla (B_0 \hat{\mathbf{z}} + \delta\mathbf{B}) &= \frac{1}{4\pi} B_0 \hat{\mathbf{z}} \cdot \nabla \delta\mathbf{B} \\ &= \frac{B_0^2}{4\pi} \nabla_\parallel \left(\delta\mathbf{b} + \frac{\delta B}{B_0} \hat{\mathbf{z}} \right)\end{aligned}$$

Combining these two terms and substituting the magnetic field strength and direction perturbations (A.7), we get

$$\begin{aligned}
& -\frac{B_0^2}{4\pi}(\nabla_{\perp} + \hat{\mathbf{z}}\nabla_{\parallel})\frac{\delta B}{B_0} + \frac{B_0^2}{4\pi}\nabla_{\parallel}\left(\delta\mathbf{b} + \frac{\delta B}{B_0}\hat{\mathbf{z}}\right) \\
& = \frac{B_0^2}{4\pi}\left(-\nabla_{\perp}\frac{\delta B}{B_0} + \nabla_{\parallel}\delta\mathbf{b}\right) \\
& = \frac{B_0^2}{4\pi}(\nabla_{\perp}\nabla_{\perp}\boldsymbol{\xi}_{\perp} + \nabla_{\parallel}^2\boldsymbol{\xi}_{\perp})
\end{aligned} \tag{A.10}$$

Assembling the terms (A.8), (A.9) and (A.10) and dividing by ρ_0 results in

$$\frac{\partial^2 \boldsymbol{\xi}}{\partial t^2} = \frac{\gamma p_0}{\rho_0} \nabla \nabla \cdot \boldsymbol{\xi} + \frac{B_0^2}{4\pi \rho_0} (\nabla_{\perp} \nabla_{\perp} \cdot \boldsymbol{\xi}_{\perp} + \nabla_{\parallel}^2 \boldsymbol{\xi}_{\perp}).$$

Defining the speed of sound c_s and the Alfvén speed v_A to be

$$c_s = \sqrt{\frac{\gamma p_0}{\rho_0}}, \quad v_A = \frac{B_0}{\sqrt{4\pi \rho_0}}$$

the equation becomes

$$\frac{\partial^2 \boldsymbol{\xi}}{\partial t^2} = c_s^2 \nabla \nabla \cdot \boldsymbol{\xi} + v_A^2 (\nabla_{\perp} \nabla_{\perp} \cdot \boldsymbol{\xi}_{\perp} + \nabla_{\parallel}^2 \boldsymbol{\xi}_{\perp}). \tag{A.11}$$

Substituting wave solutions $\boldsymbol{\xi} \propto \exp(-i\omega t + i\mathbf{k} \cdot \mathbf{r})$ into the equation above

$$\omega^2 \boldsymbol{\xi} = c_s^2 \mathbf{k} \mathbf{k} \cdot \boldsymbol{\xi} + v_A^2 (\mathbf{k}_{\perp} \mathbf{k}_{\perp} \cdot \boldsymbol{\xi}_{\perp} + k_{\parallel}^2 \boldsymbol{\xi}_{\perp}). \tag{A.12}$$

where we have defined the wavevector $\mathbf{k} = (k_{\perp}, 0, k_{\parallel})$.

Splitting the displacement field into components, we have

$$\omega^2 \xi_x = c_s^2 k_{\perp} (k_{\perp} \xi_x + k_{\parallel} \xi_{\parallel}) + v_A^2 k_{\perp}^2 \xi_x \tag{A.13}$$

$$\omega^2 \xi_y = v_A^2 k_{\parallel}^2 \xi_y \tag{A.14}$$

$$\omega^2 \xi_{\parallel} = c_s^2 k_{\parallel} (k_{\perp} \xi_x + k_{\parallel} \xi_{\parallel}) \tag{A.15}$$

The perturbations of the rest of the fields are found by substituting the wave solution into (A.5), (A.6) and (A.7).

$$\begin{aligned}
\frac{\delta \rho}{\rho_0} &= -i\mathbf{k} \cdot \boldsymbol{\xi} = -i(k_{\perp} \xi_x + k_{\parallel} \xi_{\parallel}), \\
\frac{\delta p}{p_0} &= \gamma \frac{\delta \rho}{\rho_0}, \\
\delta \mathbf{b} &= ik_{\parallel} \boldsymbol{\xi}_{\perp} = ik_{\parallel} \begin{pmatrix} \xi_x \\ \xi_y \\ 0 \end{pmatrix}, \\
\frac{\delta B}{B_0} &= -ik_{\perp} \xi_x.
\end{aligned} \tag{A.16}$$

A.3 Derivation of Alfvén Waves

We notice that equation (A.14) is in the form of an eigenvector equation with eigenvector $\boldsymbol{\xi} = (0, \xi_y, 0)$ and eigenvalues,

$$\omega = \pm k_{\parallel} v_A$$

These represent Alfvén waves that propagate parallel and antiparallel to $k_{\parallel}$ or in the $\hat{\mathbf{z}}$ direction. The value of $\mathbf{k}_{\perp}$ can be non zero, and it causes the wave profile to shift perpendicular to the propagation direction but does not affect the propagation itself.

Examining the perturbations of the fields that involve displacement in the y direction

$$\boldsymbol{\xi} = \xi_y \hat{\mathbf{y}}, \quad \delta\rho = 0 \quad \delta p = 0, \quad \delta B = 0, \quad \delta\mathbf{b} = ik_{\parallel} \xi_y \hat{\mathbf{y}}$$

we see that the curvature force of field lines is the only source of waves. Hence Alfvén waves behave similarly to waves on a string, where the strings are magnetic field lines. No compression of the fluid occurs.

A.4 Derivation of Magnetosonic Waves

Magnetosonic waves involve both sound and Alfvén speeds, therefore the equations for the x and z components of the displacement field represent magnetosonic waves. We can rewrite (A.13) and (A.15) as an eigenvector equation with the corresponding displacement field components as the eigenvector.

$$\omega^2 \begin{pmatrix} \xi_x \\ \xi_{\parallel} \end{pmatrix} = \begin{pmatrix} c_s^2 k_{\perp}^2 + v_A^2 k^2 & c_s^2 k_{\parallel} k_{\perp} \\ c_s^2 k_{\parallel} k_{\perp} & c_s^2 k_{\parallel}^2 \end{pmatrix} \cdot \begin{pmatrix} \xi_x \\ \xi_{\parallel} \end{pmatrix}$$

We can find the dispersion relation by utilising the fact that the eigenvector equation must satisfy the characteristic equation $\det(A - \omega^2 I) = 0$, where A is the matrix and I is the identity matrix.

$$\omega^4 - k^2 (c_s^2 + v_A^2) \omega^2 + c_s^2 v_A^2 k^2 k_{\parallel}^2 = 0$$

We can solve the dispersion relation using the quadratic formula to obtain

$$\omega^2 = \frac{1}{2} k^2 \left[c_s^2 + v_A^2 \pm \sqrt{(c_s^2 + v_A^2)^2 - 4c_s^2 v_A^2 \cos^2 \theta} \right] \quad (\text{A.17})$$

where we have used the substitution $\cos \theta = k_{\parallel}/k$. Fast magnetosonic waves are the high frequency solutions, whereas slow magnetosonic waves have a lower frequency. In the above expression, the upper sign represents the fast magnetosonic frequencies.

Plasma beta β , defined as the thermal and magnetic energy ratio, also represents the relative importance of the Alfvén and sound speeds in physical system for magnetosonic waves.

$$\beta = \frac{p_0}{B_0^2/8\pi} = \frac{2}{\gamma} \frac{c_s^2}{v_A^2}$$

Due to the complexity of the dispersion relation, analysing specific limiting cases are easier to undertake.

A.4.1 Parallel Propagation: $k_\perp = 0$

For parallel propagation, $k_\perp = 0$ so the eigenvalue equation becomes

$$\omega^2 \begin{pmatrix} \xi_x \\ \xi_\parallel \end{pmatrix} = \begin{pmatrix} v_A^2 k_\parallel^2 & 0 \\ 0 & c_s^2 k_\parallel^2 \end{pmatrix} \cdot \begin{pmatrix} \xi_x \\ \xi_\parallel \end{pmatrix}.$$

Since the matrix is diagonal, we observe that there are two orthogonal eigenvectors

For the eigenvector $(\xi_x, 0, 0)$, the corresponding eigenvalue is

$$\omega = \pm k_\parallel v_A.$$

This represents an Alfvén wave with displacements in the x direction, which is physically identical to the Alfvén waves produced from displacements in the y direction.

The curvature force is the only source for Alfvén waves, as expected.

$$\boldsymbol{\xi} = \xi_x \hat{\mathbf{x}}, \quad \delta\rho = 0 \quad \delta p = 0, \quad \delta B = 0, \quad \delta\mathbf{b} = ik_\parallel \xi_x \hat{\mathbf{x}}$$

The eigenvalue for eigenvector $(0, 0, \xi_\parallel)$ is

$$\omega = \pm k_\parallel c_s,$$

which represents a sound wave with compressions and rarefactions in the fluid only.

No perturbation of the magnetic field in the parallel propagation direction can occur since this perturbation only translates the magnetic field lines.

$$\boldsymbol{\xi} = \xi_\parallel \hat{\mathbf{z}}, \quad \frac{\delta\rho}{\rho_0} = -ik_\parallel \xi_\parallel \quad \frac{\delta p}{p_0} = \gamma \frac{\delta\rho}{\rho_0}, \quad \delta B = 0, \quad \delta\mathbf{b} = 0$$

In a high β regime, the sound wave is the fast wave and the Alfvén is the slow, since thermal energy dominates over magnetic energy. The opposite occurs in the low β regime

A.4.2 Perpendicular Propagation: $k_{\parallel} = 0$

When $k_{\parallel} = 0$, the eigenvector equation is

$$\omega^2 \begin{pmatrix} \xi_x \\ \xi_{\parallel} \end{pmatrix} = \begin{pmatrix} (c_s^2 + v_A^2)k_{\perp}^2 & 0 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} \xi_x \\ \xi_{\parallel} \end{pmatrix}$$

The eigenvector is $(\xi_x, 0, 0)$, corresponding eigenvalue given by the diagonal entry of the matrix.

$$\omega = \pm k_{\perp} \sqrt{c_s^2 + v_A^2} \quad (\text{A.18})$$

The dispersion relation for the magnetosonic wave represents a sound wave that travels at a combination of Alfvén and sound speeds. Hence, both magnetic and thermal pressures are sources for the fast (and only) magnetosonic wave. The perpendicular propagation enables the perturbation of the fluid density and the magnetic field strength.

$$\boldsymbol{\xi} = \xi_x \hat{\mathbf{x}}, \quad \frac{\delta \rho}{\rho_0} = -ik_{\perp} \xi_x, \quad \frac{\delta p}{p_0} = \gamma \frac{\delta \rho}{\rho_0}, \quad \frac{\delta B}{B_0} = -ik_{\perp} \xi_x, \quad \delta \mathbf{b} = 0$$

A.4.3 Anisotropic Perturbations $k_{\parallel} \ll k_{\perp}$

The anisotropic regime $k_{\parallel} \ll k_{\perp}$ is an important case to examine because structures in plasmas are highly elongated in the direction of the strong magnetic field ($l_{\perp} \sim 1/k_{\perp}$). However, unlike the $k_{\parallel}$ regime, perpendicular structures, corresponding to $l_{\parallel} \sim 1/k_{\parallel}$, are still present.

We begin by first factorising $(c_s^2 + v_A^2)$ from the magnetosonic dispersion relation (A.17). We then apply the anisotropic limit to the equation by Taylor expanding the square root to first order since the second term in the square root is very small.

$$\begin{aligned} \omega^2 &= \frac{1}{2} k^2 (c_s^2 + v_A^2) \left[1 \pm \sqrt{1 - \frac{4c_s^2 v_A^2}{(c_s^2 + v_A^2)^2} \frac{k_{\parallel}^2}{k^2}} \right] \\ &\approx \frac{1}{2} k^2 (c_s^2 + v_A^2) \left[1 \pm 1 \mp \frac{2c_s^2 v_A^2}{(c_s^2 + v_A^2)^2} \frac{k_{\parallel}^2}{k^2} \right] \end{aligned}$$

We obtain the same dispersion relation as the fast wave (A.18) for perpendicular propagation for the upper sign.

$$\omega = \pm k_{\perp} \sqrt{c_s^2 + v_A^2}$$

The slow magnetosonic wave is given by the lower sign.

$$\omega = \pm k_{\parallel} \frac{c_s v_A}{\sqrt{c_s^2 + v_A^2}} \quad (\text{A.19})$$

We can determine the relative importance of the perpendicular and parallel displacements by rearranging equation (A.15) and substituting the slow wave dispersion relation

$$\begin{aligned} (\omega^2 - c_s^2 k_{\parallel}^2) \xi_{\parallel} &= c_s^2 k_{\parallel} k_{\perp} \xi_x \\ \left(\frac{k_{\parallel}^2 c_s^2 v_A^2}{c_s^2 + v_A^2} - c_s^2 k_{\parallel}^2 \right) \xi_{\parallel} &= c_s^2 k_{\parallel} k_{\perp} \xi_x \\ - \frac{k_{\parallel} c_s^2}{c_s^2 + v_A^2} \xi_{\parallel} &= k_{\perp} \xi_x \end{aligned}$$

We see that the x displacement is very small compared to parallel displacements in the anisotropic regime.

$$\frac{\xi_x}{\xi_{\parallel}} = - \frac{k_{\parallel}}{k_{\perp}} \frac{c_s^2}{c_s^2 + v_A^2} \ll 1$$

Substituting the result above into the field perturbation equations (A.16) yields

$$\begin{aligned} \frac{\delta \rho}{\rho_0} &= -i (k_{\perp} \xi_x + k_{\parallel} \xi_{\parallel}) = -i \frac{v_A^2}{c_s^2 + v_A^2} k_{\parallel} \xi_{\parallel}, \\ \frac{\delta p}{p_0} &= \gamma \frac{\delta \rho}{\rho_0}, \\ \delta \mathbf{b} &= i k_{\parallel} \xi_x \hat{\mathbf{x}} = -i \frac{k_{\parallel}}{k_{\perp}} \frac{c_s^2}{c_s^2 + v_A^2} k_{\parallel} \xi_{\parallel} \hat{\mathbf{x}} \rightarrow 0, \\ \frac{\delta B}{B_0} &= -i k_{\perp} \xi_x = i \frac{c_s^2}{c_s^2 + v_A^2} k_{\parallel} \xi_{\parallel}. \end{aligned}$$

No perturbation on the magnetic field direction occurs to the lowest order of $k_{\parallel}/k_{\perp}$, hence the slow waves are sound waves/ The slow waves are pressure balance which means the compressions and rarefactions in the fluid out of phase with those in the field and the net pressure perturbation is zero.

$$\delta \left(p + \frac{B^2}{8\pi} \right) = p_0 \frac{\delta p}{p_0} + \frac{B_0^2}{4\pi} \frac{\delta B}{B_0} = \rho_0 \left(c_s^2 \frac{\delta p}{p_0} + v_A^2 \frac{\delta B}{B_0} \right) = 0.$$

In the anisotropic limit, Alfvén waves can propagate since $k_{\parallel} \neq 0$ and behave independently to $k_{\perp}$.

A.5 RMHD in Straight Field with Perpendicular Shear

A.5.1 Ordering and Potentials

Reduced MHD simplifies the MHD model by removing negligible or fast varying components leaving only the slow varying and interesting dynamics such as turbulence. To remove such components, we scaled and ordered terms on the basis of properties of the plasma and the dynamics we aim to study in our system. We assume that fields and motions are subsonic to eliminate fast motions which present itself as fast waves. Because perpendicular scales are much smaller than parallel scales in plasma, we adopt anisotropic ordering. We also continue to work in a regime where the perturbations of the fields are small compared to the background.

Since we have examined only the linear dynamics in the previous section about waves, our intention is to now examine a system where the non linear and linear dynamics are equally important. This is carried out by ordering the linear and nonlinear timescales to be on the same order. In the MHD equations, the nonlinearities are presented through the advection term $\mathbf{u} \cdot \nabla$. Thus, the rate of nonlinear interaction is $\sim ku$ and is ordered to be comparable to the frequency of Alfvén waves and slow waves.

A summary of all the ordering we will be using is given below.

$$\epsilon \sim \frac{\delta u_{\perp}}{c_s} \sim \frac{\delta u_{\parallel}}{c_s} \sim |\delta \mathbf{b}| \sim \frac{\delta B}{B_0} \sim \frac{\delta \rho}{\rho_0} \sim \frac{\delta p}{p_0} \sim \frac{\omega}{k_{\perp} c_s} \sim \frac{k_{\parallel}}{k_{\perp}} \ll 1$$

We can apply our ordering regime to find useful reformulations of particular fields in terms of potentials. Dividing the continuity equation (A.1) by the background density ρ_0 , we obtain the expanded equation

$$\left(\frac{\partial}{\partial t} + \mathbf{U} \cdot \nabla + \delta \mathbf{u}_{\perp} \cdot \nabla + \delta u_{\parallel} \hat{\mathbf{z}} \cdot \nabla \right) \frac{\delta \rho}{\rho_0} = - \left(1 + \frac{\delta \rho}{\rho_0} \right) (\nabla \cdot \delta \mathbf{u}_{\perp} + \hat{\mathbf{z}} \cdot \nabla \delta u_{\parallel})$$

By ordering the terms, we find that the perpendicular velocity field is 2D incompressible at first order. Give example of ordering: $\delta \mathbf{u}_{\perp} \cdot \nabla \sim \delta \mathbf{u}_{\perp} \mathbf{k}_{\perp}$, $\nabla \cdot \delta \mathbf{u}_{\perp} \sim \mathbf{k}_{\perp} \delta \mathbf{u}_{\perp}$, $\delta \rho / \rho_0$ are the zeroth order terms, $\partial / \partial t$, $\mathbf{U} \cdot \nabla = U \nabla_{\parallel} \sim U k_{\parallel}$, are terms is first order in ϵ and $\delta u_{\parallel} \hat{\mathbf{z}} \cdot \nabla \sim \delta u_{\parallel} \nabla_{\parallel} \sim \delta u_{\parallel} k_{\parallel}$, $\hat{\mathbf{z}} \cdot \nabla \delta u_{\parallel} \sim \nabla_{\parallel} \delta u_{\parallel} \sim k_{\parallel} \delta u_{\parallel}$ are second order terms in the equation, thus continuity equation at first order is the following when we expand the equation

$$\mathcal{O}(\epsilon) : \nabla \cdot \delta \mathbf{u}_{\perp} = 0 \tag{A.20}$$

Consequently, we can write $\delta \mathbf{u}_\perp$ in terms of a stream function Φ

$$\delta \mathbf{u}_\perp = \hat{\mathbf{z}} \times \nabla_\perp \Phi \quad (\text{A.21})$$

At second order, we have

$$\mathcal{O}(\epsilon^2) : (\nabla \cdot \delta \mathbf{u})_2 = - \left(\frac{\partial}{\partial t} + \mathbf{U} \cdot \nabla + \delta \mathbf{u}_\perp \cdot \nabla \right) \frac{\delta \rho}{\rho_0} = - \frac{d}{dt} \frac{\delta \rho}{\rho_0} \quad (\text{A.22})$$

where we have defined, to lowest order,

$$\frac{d}{dt} \equiv \frac{\partial}{\partial t} + \mathbf{U} \cdot \nabla + \delta \mathbf{u}_\perp \cdot \nabla$$

Expanding the equation Maxwell's equation yields

$$\nabla \cdot \mathbf{B} = \nabla \cdot \delta \mathbf{B}_\perp + \hat{\mathbf{z}} \cdot \nabla \delta B_\parallel = 0$$

At lowest order, we obtain

$$\mathcal{O}(\epsilon) : \nabla \cdot \delta \mathbf{B}_\perp = 0 \quad (\text{A.23})$$

which suggests that we can express $\delta \mathbf{B}_\perp$ in terms of the flux function Ψ . Physically Φ is the velocity vector potential and represents streamlines whereas Ψ is the magnetic vector potential and represent magnetic flux lines.

$$\frac{\delta \mathbf{B}_\perp}{\sqrt{4\pi\rho_0}} = \hat{\mathbf{z}} \times \nabla_\perp \Psi \quad (\text{A.24})$$

Below are identities which will be used in the following sections. They have been defined up to first order.

$$\frac{d}{dt} \equiv \frac{\partial}{\partial t} + \mathbf{U} \cdot \nabla + \delta \mathbf{u}_\perp \cdot \nabla \quad (\text{A.25})$$

$$\hat{\mathbf{b}}_T \cdot \nabla \equiv \hat{\mathbf{z}} \cdot \nabla + \frac{\delta \mathbf{B}_\perp}{B_0} \cdot \nabla \quad (\text{A.26})$$

$$\kappa_U^\perp \equiv \nabla_\perp \ln(U) \equiv \frac{\nabla_\perp U}{U} \quad (\text{A.27})$$

Identities (A.25) and (A.26) are the convection derivative and the advection by magnetic fields respectively. (A.27) represents the scale of the background velocity shear compared to the background velocity.

We now determine the Alfvénic and compressive perturbations that produce Alfvén waves and slow waves, respectively, and briefly examine the often forgotten entropy mode.

A.5.2 Alfvénic Perturbations

We now derive the evolution of perturbations perpendicular to the field from the induction and momentum equations only to second order. When we divide the induction equation (A.3) by the background magnetic field B_0 , we obtain the following.

$$\begin{aligned}\frac{\partial \mathbf{B}}{\partial t} &= -\mathbf{u} \cdot \nabla \mathbf{B} + \mathbf{B} \cdot \nabla \mathbf{u} - \mathbf{B} \nabla \cdot \mathbf{u} + \mathbf{u} \nabla \cdot \mathbf{B} \\ \left(\frac{\partial}{\partial t} + \mathbf{u} \cdot \nabla \right) \frac{\mathbf{B}}{B_0} &= \frac{\mathbf{B}}{B_0} \cdot \nabla \mathbf{u} - \frac{\mathbf{B}}{B_0} \nabla \cdot \mathbf{u} \\ \frac{d}{dt} \frac{\delta \mathbf{B}}{B_0} &= \mathbf{b} \cdot \nabla \mathbf{u} - \mathbf{b} \nabla \cdot \delta \mathbf{u}\end{aligned}$$

We approximate $\mathbf{b} \approx \hat{\mathbf{z}}$ in the last term since the background field dominates over the perturbations.

$$\frac{d}{dt} \frac{\delta \mathbf{B}}{B_0} = \mathbf{b} \cdot \nabla \mathbf{u} - \hat{\mathbf{z}} \nabla \cdot \delta \mathbf{u} \quad (\text{A.28})$$

We subsequently project the equation on the perpendicular plane and find to second order the following.

$$\frac{d}{dt} \frac{\delta \mathbf{B}_\perp}{B_0} = \hat{\mathbf{b}}_T \cdot \nabla \delta \mathbf{u}_\perp \quad (\text{A.29})$$

To write the equation in terms of the stream and flux function, we recognise that the convective derivative of any function f can be represented, to leading order, by the following by substituting the stream function (A.21).

$$\begin{aligned}\frac{df}{dt} &= \frac{\partial f}{\partial t} + \delta \mathbf{u}_\perp \cdot \nabla_\perp f + \mathbf{U} \cdot \nabla f \\ &= \frac{\partial f}{\partial t} + \hat{\mathbf{z}} \cdot (\nabla_\perp \Phi \times \nabla_\perp f) + \mathbf{U} \cdot \nabla f \\ &= \frac{\partial f}{\partial t} + \{\Phi, f\} + \mathbf{U} \cdot \nabla f\end{aligned} \quad (\text{A.30})$$

The poisson bracket is defined as

$$\{\Phi, f\} = \frac{\partial \Phi}{\partial x} \frac{\partial f}{\partial y} - \frac{\partial \Phi}{\partial y} \frac{\partial f}{\partial x}$$

Substituting the flux function (A.24) into the expression for advection of any function f , by the magnetic field, yields a similar equation.

$$\begin{aligned}\hat{\mathbf{b}}_T \cdot \nabla f &= \frac{\partial f}{\partial z} + \delta \mathbf{b} \cdot \nabla_\perp f \\ &= \frac{\partial f}{\partial z} + \frac{1}{v_A} \hat{\mathbf{z}} \cdot (\nabla_\perp \Psi \times \nabla_\perp f) \\ &= \frac{\partial f}{\partial z} + \frac{1}{v_A} \{\Psi, f\}\end{aligned} \quad (\text{A.31})$$

Coming back to the perpendicular induction equation (A.29), we rewrite B_0 in terms of the Alfvén speed, then substitute the flux function on the LHS of the equation. On the RHS, we substitute the stream function.

$$\begin{aligned}\frac{d}{dt} \frac{1}{v_A} \frac{\delta \mathbf{B}_\perp}{\sqrt{4\pi\rho_0}} &= \hat{\mathbf{b}}_T \cdot \nabla \delta \mathbf{u}_\perp \\ \frac{1}{v_A} \frac{d}{dt} (\hat{\mathbf{z}} \times \nabla_\perp \Psi) &= \hat{\mathbf{b}}_T \cdot \nabla (\hat{\mathbf{z}} \times \nabla_\perp \Phi)\end{aligned}$$

Using (A.30) and (A.31) and multiplying by v_A yields

$$\begin{aligned}\frac{\partial}{\partial t} (\hat{\mathbf{z}} \times \nabla_\perp \Psi) + \{\Phi, \hat{\mathbf{z}} \times \nabla_\perp \Psi\} + \mathbf{U} \cdot \nabla (\hat{\mathbf{z}} \times \nabla_\perp \Psi) &= \{\Psi, \hat{\mathbf{z}} \times \nabla_\perp \Phi\} + v_A \frac{\partial}{\partial z} (\hat{\mathbf{z}} \times \nabla_\perp \Phi) \\ \left(\hat{\mathbf{z}} \times \nabla_\perp \frac{\partial \Psi}{\partial t} \right) + \{\Phi, \hat{\mathbf{z}} \times \nabla_\perp \Psi\} - \{\Psi, \hat{\mathbf{z}} \times \nabla_\perp \Phi\} + \hat{\mathbf{z}} \times \nabla_\perp (\mathbf{U} \cdot \nabla \Psi) &= v_A \left(\hat{\mathbf{z}} \times \nabla_\perp \frac{\partial \Phi}{\partial z} \right)\end{aligned}$$

Taking the cross product and perpendicular derivative out of the Poisson brackets can be achieved with less effort by working backwards.

$$\hat{\mathbf{z}} \times \nabla_\perp \frac{\partial \Psi}{\partial t} + \hat{\mathbf{z}} \times \nabla_\perp \{\Phi, \Psi\} + \hat{\mathbf{z}} \times \nabla_\perp (\mathbf{U} \cdot \nabla \Psi) = \hat{\mathbf{z}} \times v_A \nabla_\perp \frac{\partial \Phi}{\partial z}$$

Uncrossing $\hat{\mathbf{z}}$ and integrating over the perpendicular direction (to remove $\nabla_\perp$), we obtain an RMHD equation

$$\begin{aligned}\frac{\partial \Psi}{\partial t} + \mathbf{U} \cdot \nabla \Psi + \{\Phi, \Psi\} &= v_A \frac{\partial \Phi}{\partial z} \\ \frac{d\Psi}{dt} &= v_A \frac{\partial \Phi}{\partial z}\end{aligned}\tag{A.32}$$

Dividing the momentum equation (A.2) by $\rho \approx \rho_0$ and ignoring all third order contributions we obtain

$$\frac{d\mathbf{u}}{dt} = \frac{1}{\rho_0} \left[-\nabla \left(p + \frac{B^2}{8\pi} \right) + \frac{\mathbf{B} \cdot \nabla \delta \mathbf{B}}{4\pi} \right]\tag{A.33}$$

We take the perpendicular part of the momentum equation, remove terms third order and higher and simplify by replacing ρ_0 using the sound and Alfvén speed.

$$\begin{aligned}\frac{d\delta \mathbf{u}_\perp}{dt} &= \frac{1}{\rho_0} \left[-\nabla_\perp \left(p + \frac{B^2}{8\pi} \right) + \frac{\mathbf{B} \cdot \nabla \delta \mathbf{B}_\perp}{4\pi} \right] \\ \frac{d\delta \mathbf{u}_\perp}{dt} &= -\nabla_\perp \left(\frac{c_s^2}{\gamma} \frac{\delta p}{p_0} + v_A^2 \frac{\delta B}{B_0} \right) + v_A^2 \hat{\mathbf{b}}_T \cdot \nabla \frac{\delta B_\perp}{B_0}\end{aligned}\tag{A.34}$$

At lowest order, we obtain pressure balance

$$\mathcal{O}(\epsilon) : \nabla_\perp \left(\frac{c_s^2}{\gamma} \frac{\delta p}{p_0} + v_A^2 \frac{\delta B}{B_0} \right) = 0$$

Thus,

$$\frac{\delta p}{p_0} = -\gamma \frac{v_A^2}{c_s^2} \frac{\delta B}{B_0} \quad (\text{A.35})$$

We take the perpendicular curl of the equation (A.34) to remove the second order contribution from the total pressure, attaining,

$$\mathcal{O}(\epsilon^2) : \nabla_{\perp} \times \frac{d\delta \mathbf{u}_{\perp}}{dt} = v_A^2 \nabla_{\perp} \times \left(\hat{\mathbf{b}}_T \cdot \nabla \frac{\delta B_{\perp}}{B_0} \right)$$

Substituting the perpendicular perturbations for the stream and flux functions, (A.21) and (A.24) respectively, then expanding the triple cross products yields

$$\frac{d}{dt} \nabla_{\perp}^2 \Phi = v_A \mathbf{b} \cdot \nabla \nabla_{\perp}^2 \Psi,$$

This can be rewritten as the following using (A.30) and (A.31).

$$\frac{\partial}{\partial t} \nabla_{\perp}^2 \Phi + \{\Phi, \nabla_{\perp}^2 \Phi\} + \mathbf{U} \cdot \nabla \nabla_{\perp}^2 \Phi = v_A \frac{\partial}{\partial z} \nabla_{\perp}^2 \Psi + \{\Psi, \nabla_{\perp}^2 \Psi\} \quad (\text{A.36})$$

The RMHD equations for the Alfvénic perturbations are summarised below.

$$\begin{aligned} \frac{d}{dt} \nabla_{\perp}^2 \Phi &= v_A \mathbf{b} \cdot \nabla \nabla_{\perp}^2 \Psi, \\ \frac{d\Psi}{dt} &= v_A \frac{\partial \Phi}{\partial z} \end{aligned} \quad (\text{A.37})$$

A.5.3 Compressive Perturbations

Taking the parallel component of the simplified (maybe change word) momentum equation (A.33)

$$\frac{d(U + \delta u_{\parallel})}{dt} = \frac{1}{\rho_0} \left[-\nabla_{\parallel} \left(p + \frac{B^2}{8\pi} \right) + \frac{\mathbf{B} \cdot \nabla \delta B_{\parallel}}{4\pi} \right] \quad (\text{A.38})$$

Since pressure balance occurs at lowest order, the first term on the right side is third order and is ignored. Expanding the term dU/dt and substituting an expression for ρ_0 in terms of the Alfvén speed, we obtain

$$\frac{d\delta u_{\parallel}}{dt} = v_A^2 \hat{\mathbf{b}}_T \cdot \nabla \frac{\delta B_{\parallel}}{B_0} - U \delta \mathbf{u}_{\perp} \cdot \boldsymbol{\kappa}_{\mathbf{U}}^{\perp} \quad (\text{A.39})$$

Projecting the simplified induction equation (A.28) along the parallel direction, we obtain

$$\frac{d}{dt} \frac{\delta B}{B_0} = \hat{\mathbf{b}}_T \cdot \nabla (U + \delta u_{\parallel}) - \nabla \cdot \delta \mathbf{u} \quad (\text{A.40})$$

where $\delta B/B_0 = \delta B_{\parallel}/B_0$. Substituting (A.22) into the last term and expanding $\hat{\mathbf{b}}_T \cdot \nabla U$, we obtain

$$\frac{d}{dt} \left(\frac{\delta B_{\parallel}}{B_0} - \frac{\delta \rho}{\rho_0} \right) = \hat{\mathbf{b}}_T \cdot \nabla \delta u_{\parallel} + U \frac{\delta \mathbf{B}_{\perp}}{B_0} \cdot \boldsymbol{\kappa}_U^{\perp} \quad (\text{A.41})$$

To write the above equation only in terms of the velocity and magnetic field, we turn to the energy equation (A.4). First, dividing by s_0 , we then expand the equation in terms of p and ρ

$$\frac{d}{dt} \frac{\delta s}{s_0} = \frac{d}{dt} \left(\frac{\delta p}{p_0} - \gamma \frac{\delta \rho}{\rho_0} \right) = 0 \quad (\text{A.42})$$

Substituting the pressure balance (A.35), we find the equation in second order to be

$$\frac{v_A^2}{c_s^2} \frac{d}{dt} \frac{\delta B_{\parallel}}{B_0} = - \frac{d}{dt} \frac{\delta \rho}{\rho_0} \quad (\text{A.43})$$

Finally, we substitute (A.43) into (A.41).

$$\left(1 + \frac{v_A^2}{c_s^2} \right) \frac{d}{dt} \frac{\delta B_{\parallel}}{B_0} = \hat{\mathbf{b}}_T \cdot \nabla \delta u_{\parallel} + U \frac{\delta \mathbf{B}_{\perp}}{B_0} \cdot \boldsymbol{\kappa}_U^{\perp} \quad (\text{A.44})$$

The RMHD equations for the compressive perturbations are summarised below.

$$\begin{aligned} \frac{d \delta u_{\parallel}}{dt} &= v_A^2 \hat{\mathbf{b}}_T \cdot \nabla \frac{\delta B_{\parallel}}{B_0} - U \delta \mathbf{u}_{\perp} \cdot \boldsymbol{\kappa}_U^{\perp} \\ \left(1 + \frac{v_A^2}{c_s^2} \right) \frac{d}{dt} \frac{\delta B_{\parallel}}{B_0} &= \hat{\mathbf{b}}_T \cdot \nabla \delta u_{\parallel} + U \frac{\delta \mathbf{B}_{\perp}}{B_0} \cdot \boldsymbol{\kappa}_U^{\perp} \end{aligned} \quad (\text{A.45})$$

A.5.4 Entropy Mode

The entropy equation (A.4) also describes a perturbation, the entropy perturbation.

$$\frac{d \delta s}{dt} = 0 \quad (\text{A.46})$$

The perturbation δs is passively advected by the background flow $\mathbf{U}$ and the perpendicular velocity perturbation $\delta \mathbf{u}_{\perp}$. As a result, no waves are produced from entropy perturbations.

Since s is dependent on density, the entropy perturbation is decoupled from the Alfvénic and slow perturbations.

A.5.5 Elsasser Formulation and Energy

Elsasser Fields

Due to the underlying symmetry between perpendicular velocity and magnetic field perturbations it is helpful to recast them into Elsasser field $\delta \mathbf{Z}_{\perp}^{\pm}$.

$$\delta \mathbf{Z}_{\perp}^{\pm} = \mathbf{u}_{\perp} \mp \frac{\delta \mathbf{B}_{\perp}}{\sqrt{4\pi\rho_0}} \quad (\text{A.47})$$

The upper sign in the equation above represents outward propagating Alfvénic perturbations and the lower sign represents Alfvénic perturbations traveling inwards.

The Elsasser fields can be represented as the following using the potential formulation for the perpendicular perturbations, (A.21) and (A.24)

$$\delta \mathbf{Z}_{\perp}^{\pm} = \hat{\mathbf{z}} \times \nabla_{\perp} \zeta^{\pm},$$

where we have defined the Elsasser potentials as

$$\zeta^{\pm} = \Phi \mp \Psi.$$

The ‘vorticities’ of the Elsasser fields, which physically represent fluid vorticity (stream) and electrical currents (flux), is defined as

$$\omega^{\pm} = \hat{\mathbf{z}} \cdot (\nabla_{\perp} \times \delta \mathbf{Z}_{\perp}^{\pm}) = \nabla_{\perp}^2 \zeta^{\pm}. \quad (\text{A.48})$$

We can rewrite the Alfvénic RMHD equations (A.37) into a evolution equation for the ‘vorticities’. Firstly, we expand the equation using (A.30), then multiply the RMHD equation (A.32) by $\nabla_{\perp}^2$.

$$\begin{aligned} \frac{\partial}{\partial t}(\nabla_{\perp}^2 \Psi) + \nabla_{\perp}^2 \{\Phi, \Psi\} + \mathbf{U} \cdot \nabla(\nabla_{\perp}^2 \Psi) &= v_A \frac{\partial}{\partial z}(\nabla_{\perp}^2 \Phi) \\ \frac{\partial}{\partial t}(\nabla_{\perp}^2 \Psi) + \{\nabla_{\perp}^2 \Phi, \Psi\} + \{\Phi, \nabla_{\perp}^2 \Psi\} + \mathbf{U} \cdot \nabla(\nabla_{\perp}^2 \Psi) &= v_A \frac{\partial}{\partial z}(\nabla_{\perp}^2 \Phi) \end{aligned}$$

Subtracting (adding) the above from (to) the RMHD equation (A.36), results in

$$\begin{aligned} \frac{\partial \omega^{\pm}}{\partial t} \pm v_A \frac{\partial \omega^{\pm}}{\partial z} + \{\Phi, \nabla_{\perp}^2 \Phi\} - \{\Psi, \nabla_{\perp}^2 \Psi\} \mp \{\nabla_{\perp}^2 \Phi, \Psi\} \mp \{\Phi, \nabla_{\perp}^2 \Psi\} + \mathbf{U} \cdot \nabla \omega^{\pm} &= 0 \\ \frac{\partial \omega^{\pm}}{\partial t} \pm v_A \frac{\partial \omega^{\pm}}{\partial z} + \{\Phi, \omega^{\pm}\} \pm \{\Psi, \omega^{\pm}\} + \mathbf{U} \cdot \nabla \omega^{\pm} &= 0 \\ \frac{\partial \omega^{\pm}}{\partial t} + (U \pm v_A) \frac{\partial \omega^{\pm}}{\partial z} + \{\zeta^{\mp}, \omega^{\pm}\} &= 0 \\ \frac{\partial \omega^{\pm}}{\partial t} + (U \pm v_A) \frac{\partial \omega^{\pm}}{\partial z} + \{\zeta^{\mp}, \omega^{\pm}\} &= \{\partial_j \zeta^{\pm}, \partial_j \zeta^{\mp}\} \end{aligned} \quad (\text{A.49})$$

Spil later about what RHS represents

Counterpropagating Alfvénic perturbations, propagating with speed $U \pm v_A$ can only interact nonlinearly with each other (Alfvén waves traveling in one direction have the same speed). A perturbation composed of a single Elsasser field is a nonlinear solution to the equation. For example, if $\delta \mathbf{Z}_{\perp}^{-} = 0$, then the evolution equation becomes

$$\frac{\partial \omega^{+}}{\partial t} + (U \pm v_A) \frac{\partial \omega^{+}}{\partial z} = 0$$

and ω^+ and therefore $\delta \mathbf{Z}_\perp^+$ is a solution.

There is an Elsasser-like formulation for the slow waves as well, where again the upper sign represents outward propagating waves and the lower sign, inwards.

$$\delta Z_\parallel^\pm = u_\parallel \mp \frac{\delta B}{\sqrt{4\pi\rho_0}} \sqrt{1 + \frac{v_A^2}{c_s^2}} \quad (\text{A.50})$$

First we convert the parallel magnetic field perturbation in the RMHD equations (A.45), into the form required for Elsasser-like formulation.

$$\begin{aligned} \frac{d\delta u_\parallel}{dt} &= \frac{v_A c_s}{\sqrt{v_A^2 + c_s^2}} \hat{\mathbf{b}}_T \cdot \nabla \frac{\delta B_\parallel}{\sqrt{4\pi\rho_0}} \sqrt{1 + \frac{v_A^2}{c_s^2}} - U \delta \mathbf{u}_\perp \cdot \boldsymbol{\kappa}_U^\perp \\ \frac{d}{dt} \frac{\delta B_\parallel}{\sqrt{4\pi\rho_0}} \sqrt{1 + \frac{v_A^2}{c_s^2}} &= \frac{v_A c_s}{\sqrt{v_A^2 + c_s^2}} \hat{\mathbf{b}}_T \cdot \nabla \frac{\delta u_\parallel}{\sqrt{4\pi\rho_0}} \sqrt{1 + \frac{v_A^2}{c_s^2}} + U \frac{c_s}{\sqrt{v_A^2 + c_s^2}} \frac{\delta \mathbf{B}_\perp}{\sqrt{4\pi\rho_0}} \cdot \boldsymbol{\kappa}_U^\perp \end{aligned}$$

Subtracting (adding) the first equation above from (to) the second equation above and then expanding using (A.30) and (A.31), we obtain

$$\begin{aligned} \frac{d\delta Z_\parallel^\pm}{dt} \pm \frac{c_s v_A}{\sqrt{c_s^2 + v_A^2}} \hat{\mathbf{b}}_T \cdot \nabla \delta Z_\parallel^\pm &= -U \boldsymbol{\kappa}_U^\perp \cdot \left(\delta \mathbf{u}_\perp \pm \frac{c_s}{\sqrt{v_A^2 + c_s^2}} \frac{\delta \mathbf{B}_\perp}{\sqrt{4\pi\rho_0}} \right) \\ \frac{\partial \delta Z_\parallel^\pm}{\partial t} + \left(U \pm \frac{c_s v_A}{\sqrt{c_s^2 + v_A^2}} \right) \frac{\partial \delta Z_\parallel^\pm}{\partial z} + \{\Phi, \delta Z_\parallel^\pm\} \pm \frac{c_s}{\sqrt{c_s^2 + v_A^2}} \{\Psi, \delta Z_\parallel^\pm\} \\ &= -U \boldsymbol{\kappa}_U^\perp \cdot \left(\delta \mathbf{u}_\perp \pm \frac{c_s}{\sqrt{v_A^2 + c_s^2}} \frac{\delta \mathbf{B}_\perp}{\sqrt{4\pi\rho_0}} \right) \end{aligned}$$

Utilising the fact that $2\Phi = \zeta^+ + \zeta^-$ and $2\Psi = \zeta^+ - \zeta^-$, the equation becomes

$$\begin{aligned} \frac{\partial \delta Z_\parallel^\pm}{\partial t} + \left(U \pm \frac{c_s v_A}{\sqrt{c_s^2 + v_A^2}} \right) \frac{\partial \delta Z_\parallel^\pm}{\partial z} \\ = -\frac{1}{2} \left[\left(1 \pm \frac{1}{\alpha} \right) [\{\zeta^+, \delta Z_\parallel^\pm\} + U \boldsymbol{\kappa}_U^\perp \cdot \delta \mathbf{Z}_\perp^+] + \left(1 \mp \frac{1}{\alpha} \right) [\{\zeta^-, \delta Z_\parallel^\pm\} + U \boldsymbol{\kappa}_U^\perp \cdot \delta \mathbf{Z}_\perp^-] \right], \end{aligned} \quad (\text{A.51})$$

where $\alpha = \sqrt{1 + v_A^2/c_s^2}$.

The evolution equation represents slow waves traveling at $U \pm (c_s v_A)/\sqrt{c_s^2 + v_A^2}$, the background velocity $\pm$ slow wave speed. When the Alfvén speed is much slower than the sound speed, $v_A \ll c_s$ (subsonic), slow waves only interact with counterpropagating Alfvén perturbations. However, when $v_A \sim c_s$ or higher, slow waves can interact with copropagating Alfvén waves since these Alfvén waves can pass the slow waves traveling

in the same direction. Slow waves are also driven only by counterpropagating Alfvén waves when $v_A \ll c_s$ and can be driven by both counter and copropagating Alfvén waves in the presence of the background velocity shear. **Causality-because slow waves can only interact with counterpropagating Alfvén waves, these waves can be the only ones that drive the slow waves (use energy from the velocity shear to drive them)**

In the next section we determine whether energies of $\delta \mathbf{Z}_\perp^\pm$ and $\delta Z_\parallel^\pm$ are conserved or not.

Alfvén Wave Energy

The energy of the Alfvén waves, stored in the Elsasser fields $\delta \mathbf{Z}_\perp^\pm$, is given by

$$E_{\text{Alf}} = \frac{1}{2} \int dV \rho_0 |\delta \mathbf{Z}_\perp^\pm|^2.$$

Rewriting in terms of the perpendicular velocity and magnetic field, the Alfvén energy becomes

$$\begin{aligned} E_{\text{Alf}} &= \frac{1}{2} \int dV \rho_0 \frac{|\delta \mathbf{Z}_\perp^+|^2 + |\delta \mathbf{Z}_\perp^-|^2}{2} \\ &= \int dV \frac{\rho_0}{2} \delta \mathbf{u}_\perp^2 + \frac{\delta \mathbf{B}_\perp^2}{8\pi}. \end{aligned}$$

The Alfvénic energy has kinetic and magnetic energy contributions.

By taking the time derivative of the Alfvén wave energy, we can see how the energy of the waves evolves.

$$\frac{\partial E_{\text{Alf}}}{\partial t} = \int dV \frac{\partial}{\partial t} \left(\frac{\rho_0}{2} \delta \mathbf{u}_\perp^2 \right) + \frac{\partial}{\partial t} \left(\frac{\delta \mathbf{B}_\perp^2}{8\pi} \right).$$

If we convert the terms in the integral into a total divergence, we can remove them from the equation since there are no sources or sinks of energy within a fluid element of plasma.

We first evaluate the time derivative of the kinetic energy term in the integral by substituting the equation (A.5.2).

$$\begin{aligned} \frac{\partial}{\partial t} \frac{\rho_0}{2} \delta \mathbf{u}_\perp^2 &= \delta \mathbf{u}_\perp \cdot \left(\rho_0 \frac{\partial}{\partial t} \delta \mathbf{u}_\perp \right) \\ &= -\rho_0 \delta \mathbf{u}_\perp \cdot (\tilde{\mathbf{u}} \cdot \nabla \delta \mathbf{u}_\perp) - \delta \mathbf{u}_\perp \cdot \nabla \left(p + \frac{B^2}{8\pi} \right) + \delta \mathbf{u}_\perp \cdot \frac{\mathbf{B} \cdot \nabla \delta \mathbf{B}_\perp}{4\pi} \end{aligned} \quad (\text{A.52})$$

We have defined $\tilde{\mathbf{u}} = \mathbf{U} + \delta \mathbf{u}_\perp$ in the above.

We convert all the terms into divergences, beginning with the first term

$$\begin{aligned} -\rho_0 \delta \mathbf{u}_\perp \cdot (\tilde{\mathbf{u}} \cdot \nabla \delta \mathbf{u}_\perp) &= -\frac{\rho_0}{2} \tilde{\mathbf{u}} \cdot \nabla |\delta \mathbf{u}_\perp|^2 \\ &= -\nabla \cdot \left(\frac{\rho_0}{2} \delta u_\perp^2 \tilde{\mathbf{u}} \right) \end{aligned}$$

where we have used the fact that $\nabla \cdot \tilde{\mathbf{u}} = 0$ (to first order).

The second term in (A.52) becomes

$$-\nabla \cdot \left[\left(p + \frac{B^2}{8\pi} \right) \cdot \delta \mathbf{u}_\perp \right]$$

The result for the final term also contains an extra non divergence term.

$$\begin{aligned} \frac{1}{4\pi} \delta \mathbf{u}_\perp \cdot \mathbf{B} \cdot \nabla \delta \mathbf{B}_\perp &= \frac{1}{4\pi} \delta \mathbf{u}_\perp \cdot \nabla (\delta \mathbf{B}_\perp \mathbf{B}) \\ &= \nabla \cdot \left(\frac{1}{4\pi} \delta \mathbf{u}_\perp \cdot \delta \mathbf{B}_\perp \mathbf{B} \right) - \frac{\delta \mathbf{B}_\perp}{4\pi} \cdot (\mathbf{B} \cdot \nabla \delta \mathbf{u}_\perp) \end{aligned}$$

For the time derivative of the magnetic energy, we substitute the equation (A.29).

$$\begin{aligned} \frac{\partial}{\partial t} \left(\frac{\delta \mathbf{B}_\perp^2}{8\pi} \right) &= \frac{\delta \mathbf{B}_\perp}{4\pi} \cdot \frac{\partial \delta \mathbf{B}_\perp}{\partial t} \\ &= -\frac{1}{4\pi} \delta \mathbf{B}_\perp \cdot (\tilde{\mathbf{u}} \cdot \nabla \delta \mathbf{B}_\perp) + \frac{1}{4\pi} \delta \mathbf{B}_\perp \cdot (\mathbf{B} \cdot \nabla \delta \mathbf{u}_\perp) \end{aligned} \tag{A.53}$$

We can convert the first term in the equation above into a divergence.

$$\begin{aligned} -\frac{1}{4\pi} \delta \mathbf{B}_\perp \cdot (\tilde{\mathbf{u}} \cdot \nabla \delta \mathbf{B}_\perp) &= -\tilde{\mathbf{u}} \cdot \nabla \frac{\delta B_\perp^2}{8\pi} \\ &= -\nabla \cdot \left(\frac{\delta B_\perp^2}{8\pi} \tilde{\mathbf{u}} \right) \end{aligned}$$

The the kinetic and magnetic energy terms are substituted into the integral and the expression is simplified.

$$\frac{\partial E_{\text{Alf}}}{\partial t} = \int \left[-\nabla \cdot \left(\frac{\rho_0}{2} \delta u_\perp^2 \tilde{\mathbf{u}} + \left(p + \frac{B^2}{8\pi} \right) \cdot \delta \mathbf{u}_\perp + \frac{1}{4\pi} \delta \mathbf{u}_\perp \cdot (\delta \mathbf{B}_\perp \mathbf{B}) - \frac{\delta B_\perp^2}{8\pi} \tilde{\mathbf{u}} \right) \right] dV$$

Because the Alfvénic energy can only be transferred through a fluid element and cannot be created or destroyed, the net Alvenic energy flux in the fluid element is zero and the integral is zero.

Hence, the energy of the Alfvén waves is conserved since the time derivative of the Alfvén energy is zero.

$$\frac{d}{dt} \int d\mathbf{r}^3 |\delta \mathbf{Z}_\perp^\pm|^2 = 0 \tag{A.54}$$

Slow Wave Energy

The energy of slow waves is given by the expression below.

$$E_{\text{slow}} = \frac{1}{2} \int dV \rho_0 \left| Z_{\parallel}^{\pm} \right|^2.$$

Using the Elsasser-like formulation for slow waves (A.50), the energy becomes

$$\begin{aligned} E_{\text{Slow}} &= \frac{1}{2} \int dV \rho_0 \frac{\left| \delta Z_{\parallel}^+ \right|^2 + \left| \delta Z_{\parallel}^- \right|^2}{2} \\ &= \frac{1}{2} \int dV \frac{\rho_0}{2} \delta u_{\parallel}^2 + \frac{\delta B_{\parallel}^2}{8\pi} \left(1 + \frac{v_A^2}{c_s^2} \right). \end{aligned}$$

Apart from the kinetic and magnetic energy contributions to the slow wave energy, the the wave speed scaling term originates from the internal energy of the fluid.

Once again, we wish to determine the time evolution of the slow mode energy.

$$\frac{\partial E_{\text{Slow}}}{\partial t} = \int dV \frac{\partial}{\partial t} \left(\frac{\rho_0}{2} \delta u_{\parallel}^2 \right) + \frac{\partial}{\partial t} \frac{\delta B_{\parallel}^2}{8\pi} \left(1 + \frac{v_A^2}{c_s^2} \right).$$

Expanding the kinetic energy term using the equation (A.38), we obtain

$$\begin{aligned} \frac{\partial}{\partial t} \left(\frac{\rho_0}{2} \delta u_{\parallel}^2 \right) &= \delta u_{\parallel} \rho_0 \frac{\partial}{\partial t} \delta u_{\parallel} \\ &= -\rho_0 \delta u_{\parallel} \tilde{\mathbf{u}} \cdot \nabla \delta u_{\parallel} + \delta u_{\parallel} \frac{\mathbf{B} \cdot \nabla \delta B_{\perp}}{4\pi} - \rho_0 U \delta u_{\parallel} \delta \mathbf{u}_{\perp} \cdot \boldsymbol{\kappa}_U^{\perp} \end{aligned}$$

We can convert the first two terms into divergences to get

$$\begin{aligned} -\rho_0 \delta u_{\parallel} \tilde{\mathbf{u}} \cdot \nabla \delta u_{\parallel} &= -\frac{\rho_0}{2} \tilde{\mathbf{u}} \cdot \nabla \delta u_{\parallel}^2 \\ &= -\nabla \cdot \left(\frac{\rho_0}{2} \delta u_{\parallel}^2 \tilde{\mathbf{u}} \right) \end{aligned}$$

and

$$\begin{aligned} \frac{1}{4\pi} \delta u_{\parallel} \mathbf{B} \cdot \nabla \delta B_{\parallel} &= \frac{1}{4\pi} \delta u_{\parallel} \nabla (\delta B_{\parallel} \mathbf{B}) \\ &= \nabla \cdot \left(\frac{1}{4\pi} \delta u_{\parallel} \delta B_{\parallel} \mathbf{B} \right) - \frac{\delta B_{\parallel}}{4\pi} (\mathbf{B} \cdot \nabla \delta u_{\parallel}) \end{aligned}$$

We can substitute the equation (A.44) into the last term in the integral.

$$\begin{aligned} \alpha \frac{\partial}{\partial t} \frac{\delta B_{\parallel}^2}{8\pi} &= \alpha \frac{\delta B_{\parallel}}{4\pi} \frac{\partial \delta B_{\parallel}}{\partial t} \\ &= -\frac{\alpha}{4\pi} \delta B_{\parallel} (\tilde{\mathbf{u}} \cdot \nabla \delta B_{\parallel}) + \frac{\delta B_{\parallel}}{4\pi} (\mathbf{B} \cdot \nabla \delta u_{\parallel}) + \frac{1}{4\pi} U \delta B_{\parallel} \delta \mathbf{B}_{\perp} \cdot \boldsymbol{\kappa}_U^{\perp} \end{aligned}$$

The constant is defined as $\alpha = (1 + v_A^2/c_s^2)$.

The first term of this expression becomes

$$\begin{aligned} -\frac{\alpha}{4\pi}\delta B_{\parallel}(\tilde{\mathbf{u}} \cdot \nabla \delta B_{\parallel}) &= -\alpha \tilde{\mathbf{u}} \cdot \nabla \frac{\delta B_{\parallel}^2}{8\pi} \\ &= -\nabla \cdot \left(\alpha \frac{\delta B_{\parallel}^2}{8\pi} \tilde{\mathbf{u}} \right) \end{aligned}$$

whereas the second term cancels with the corresponding term from the kinetic energy.

Assembling all terms in the integral, we get the expression

$$\frac{\partial E_{\text{Slow}}}{\partial t} = \int \left[-\nabla \cdot \left(\frac{\rho_0}{2} \delta u_{\parallel}^2 \tilde{\mathbf{u}} - \frac{1}{4\pi} \delta u_{\parallel} \delta B_{\parallel} \mathbf{B} + \alpha \frac{\delta B_{\parallel}^2}{8\pi} \tilde{\mathbf{u}} \right) - \rho_0 U (\delta u_{\parallel} \delta \mathbf{u}_{\perp} - \frac{\delta B_{\parallel} \delta \mathbf{B}_{\perp}}{4\pi \rho_0}) \cdot \boldsymbol{\kappa}_U^{\perp} \right] dV$$

There are no sources or sinks of slow wave energy in a fluid element, therefore the divergence in the integral goes to zero.

Thus, the slow wave energy is not conserved and evolves according to the equation.

$$\frac{\partial E_{\text{Slow}}}{\partial t} = \int \left[-\rho_0 U \left(\delta u_{\parallel} \delta \mathbf{u}_{\perp} - \frac{\delta B_{\parallel} \delta \mathbf{B}_{\perp}}{4\pi \rho_0} \right) \cdot \boldsymbol{\kappa}_U^{\perp} \right] dV \quad (\text{A.55})$$

The equation physically represents the kinetic energy exchange between the background velocity shear and the slow waves. This make intuitive sense since the compressive perturbation equations (A.44) and (A.39) represent the process of Alfvén waves using the energy from the velocity shear to drive compressive perturbations

Entropy Mode Energy

Since the entropy mode is decoupled from the Alfvén and slow modes, the energy of this mode is individually conserved.

$$E_s = \frac{1}{2} \int dV \frac{\delta s^2}{s_0^2}$$

Proof: Using the time evolution of entropy (and there is a contribution from internal energy)

$$\begin{aligned} \frac{\partial E_s}{\partial t} &= \int dV \frac{1}{2} \frac{\partial}{\partial t} \frac{\delta s^2}{s_0^2} \\ &= \int dV \frac{\delta s}{s_0^2} \frac{\partial \delta s}{\partial t} \\ &= \int dV -\frac{\delta s}{s_0^2} \tilde{\mathbf{u}} \cdot \nabla \delta s \\ &= \int dV \left[-\nabla \cdot \left(\frac{\delta s^2}{s_0^2} \tilde{\mathbf{u}} \right) \right] \end{aligned}$$

The divergence goes to zero since there are no sources or sinks for the entropy mode energy in a fluid element.

Thus,

$$\frac{d}{dt} \int d\mathbf{r}^3 |\delta s|^2 = 0 \quad (\text{A.56})$$

Appendix B

Source Code for thesis.dvi

You do not need to include the source code for your thesis (this is just an example). However, including code you have written to solve numerical problems can be useful – talk to your supervisor.

B.1 thesis.tex

```
1  %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
   %% This is a document template for an Otago thesis (Masters or PhD).
   %% A skeleton chapter layout is also suggested.
   %%
   %% Look in the directory example_document for filled-out chapters
   %% that show you how to do figures, bibliographies, and tables.
   %%
   %% Since this was written for Computer Science at Otago University,
   %% Harvard (author, date) style citations are used. Other students
10  %% should consult their advisors as to what style of citations to use.
   %%
   %% This template was created by Nathan Rountree 9/2/98
   %% Updates have since been made by others in the department.
   %%
   %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

   %%
   %% In the style of a technical report, in 12pt and one sided.
   %% Start chapters on right hand side pages only.
20  %%
   \documentclass[12pt]{report}
   %%\documentclass[12pt,twoside,openright]{report} %% Use this for twosided.

   %%
   %% Load packages.
   %%
   \usepackage[bschons,nolibary]{otagothesis}      %% Use Otago page layout
   %% Use [bschons] for BScHons thesis
   %% [msc]      for MSc
30  %% [phd]      for PhD
```

```

%%      [dipsci] for PGDipSci
%%      nolibrary-option omits a library declaration form.

\usepackage[longnamesfirst,round]{natbib} %% Use Natural Sciences bibliography

\usepackage[T1]{fontenc} %% assist clipboard copy from PDF; allow < | > etc.
\usepackage[utf8]{inputenc} %% allow UTF-8 content in source files

%%\usepackage{times}                %% Times PostScript font. Don't use
40                                %% if thesis contains lots of math.

\usepackage{graphicx}              %% jpg, gif, tiff, and pdf graphics
\usepackage{moreverb}              %% Verbatim Code Listings
\usepackage{xcolor}

% Standard Physics additions
\usepackage{amssymb,amsmath}
\usepackage{siunitx}
\usepackage[colorlinks=true,pdfstartview=FitV,linkcolor=blue,
50             citecolor=blue,urlcolor=blue]{hyperref}

%%
%% Set title, author and date.
%%
\title{Thesis title here} % <-- Your thesis title here
\author{Mary Wang} % <-- Add your name here
\date{\today} % <-- Submission date here.

%%
60 %% The library want to know all sorts of personal stuff!
%% Can be left out if you don't use the \frontstuff command
%%
\fullname{Mary Wang} % <-- Add your name here
\department{Department of Physics}
%\dob{1 January 1900} %% date-of-birth, only needed for library declaration
\address{730 Cumberland Street, Dunedin, NZ}

%%
70 %% Uncomment to just print up a few chapters.
%%
%%\includeonly{literature,conclusion}

%%
%% Go!
%%
\begin{document}

%%
80 %% Put in titlepage and contents, etc...
%%
\frontstuff

%%
%% Set to one-and-a-half line-spacing
%%

```

```

\linespread{1.3} \normalsize

%%
%% Include each chapter as a separate file.
90  %% These lines assume there are files called intro.tex, literature.tex etc.
%%
\include{intro}

\include{background}

\include{methods}

\include{implementation}

100 \include{results}

\include{conclusion}

%%
%% Make certain the ‘‘references’’ section begins on a recto page when
%% document is double-sided.
%% The ‘‘bibliography’’ line assumes that there is a file called
%% ‘‘thesis.bib’’ and that somewhere in the chapter material you have
%% cited something from it.
110 %%
\cleardoublepage
\bibliographystyle{otago}
\bibliography{thesis}

\include{appendices}

\end{document}
%% All Done!

```

B.1.1 abstract.tex

1 Put the abstract in this file

B.1.2 acknowledgements.tex

1 Put the acknowledgements in this file

B.1.3 intro.tex

```

1 \chapter{Introduction}
  \label{chap:intro}

```

This is the first chapter. Here is an example citation:
\citet{routtree98}.

B.1.4 literature.tex

B.1.5 new_ideas.tex

B.1.6 implementation.tex

```
1 \chapter{Implementation}
  \label{chap:implementation}
```

B.1.7 results.tex

```
1 \chapter{Results}
  \label{chap:results}
```

B.1.8 conclusion.tex

```
1 \chapter{Conclusion}
  \label{chap:conclusion}
```